



What is Nmap?

Nmap is a free open source network scanning software tool which is used for Ethical Hacking, discovery & Cyber Security audits. Nmap discovers hosts and services on a computer network by sending packets and analyzing the responses with rich feature sets.

- Usage: Ethical Hacking, Cyber Security, Auditing
- Founder: Gordon Lyon
- Official Website: www.nmap.org
- License Type: Open Source, Free
- Limitations: Use only for Ethical purposes
- Year founded: 1997 (Initial release)
- This sheet author: Networkwalks Academy

Target Selection

SWITCH /SYNTAX	EXAMPLE
Scan a single IP	<code>nmap 10.0.0.1</code>
Scan a host	<code>nmap www.networkwalks.com</code>
Scan a range of IPs	<code>nmap 10.0.0.1-99</code>
Scan multiple of Ips	<code>nmap 10.0.0.1,10.0.0.2</code>
Scan a subnet	<code>nmap 10.0.0.0/24</code>
Scan targets from a text file	<code>nmap -iL LIST1.txt</code>
Exclude targets from scan	<code>nmap 10.0.0.0 -exclude 10.0.0.0</code>

Target Scan Options



SWITCH	EXAMPLE	DESCRIPTION
-sS	<code>nmap 10.0.0.1 -sS</code>	SYN scan (TCP SYN / Stealth scan)
-sT	<code>nmap 10.0.0.1 -sT</code>	TCP connect scan (w/o root)
-sA	<code>nmap 10.0.0.1 -sA</code>	ACK scan (TCP ACK)
-sF	<code>nmap 10.0.0.1 -sF</code>	FIN scan (TCP FIN)
-sU	<code>nmap 10.0.0.1 -sU</code>	UDP Scan
-sX	<code>nmap 10.0.0.1 -sX</code>	Xmas Scan (FIN, URG, PSH)
-sL	<code>nmap 10.0.0.1 -sL</code>	List scan (only list down)
-sP	<code>nmap 10.0.0.1 -sP</code>	Ping Scan
-traceroute	<code>nmap 10.0.0.1 -traceroute</code>	(Original Author of this sheet is: www.networkwalks.com)
-A	<code>nmap 10.0.0.1 -A</code>	Aggressive Scan
-6	<code>nmap 10.0.0.1 -6</code>	IPv6 Scan
-F	<code>nmap 10.0.0.1 -F</code>	Fast Scan

Selection of Ports to scan

SWITCH	EXAMPLE
Scan a single port 21	<code>nmap 10.0.0.1 -p 21</code>
Scan port 21 and 22	<code>nmap 10.0.0.1 -p 21,22</code> or <code>nmap 10.0.0.1 -p 21-22</code>
Scan ssh port	<code>nmap 10.0.0.1 -p ssh</code>
Scan all port	<code>nmap 10.0.0.1 -p-</code>

Miscellaneous



SWITCH	EXAMPLE
Save output to file	<code>nmap -oN outputfile.txt 10.0.0.1</code>
-S	Spoof the IP
-e	Interface
--spooof-mac	MAC Spoof
--scrip	NSE (nmap Scripting Engine)

Scan speed options



SWITCH	EXAMPLE
-T4,-T5	Faster Speed (less accurate, easily detectable)
-T3	Normal Speed
-T0,-T1,-T2	Slower Speed

Target Version scan

SW	EXAMPLE	DESCRIPTION
-O	<code>nmap 10.0.0.1 -O</code>	Operating System detection
-sV	<code>nmap 10.0.0.1 -sV</code>	Service version detection

ETHICAL HACKING SERIES



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